Homework 8

Information

Name	UNI
Zhewen Guo	zg2567

Goals

Homework:

Load the data from runs 1-4 in MATLAB. The trigger (channel 18) marks second 2 of each trial. Information about the cue direction in each run and trial is in .mat files on Courseworks in the Experiment Files/Lab 9/Homework folder. Using this information, separate the trials into two groups: left (row 1) and right (row 2). There should be 80 left and 80 right. (1 pt)

Filter your data from 8-20Hz and find the average power of each electrode during left versus right imagined movements. Use the time interval when the subject is actively imagining movement (4.5s to 8s after trial onset). Can you see a clear difference between the two? (6 pts)

Calculate the difference between the two conditions for all electrodes and display the subtraction on the scalp map. You can find the new electrode configuration file 'CSP.locs' in Courseworks. In which parts of the brain do you observe a bigger difference? (3 pts)

Figure 1. screenshot of requirements

Task 1: Load dataset

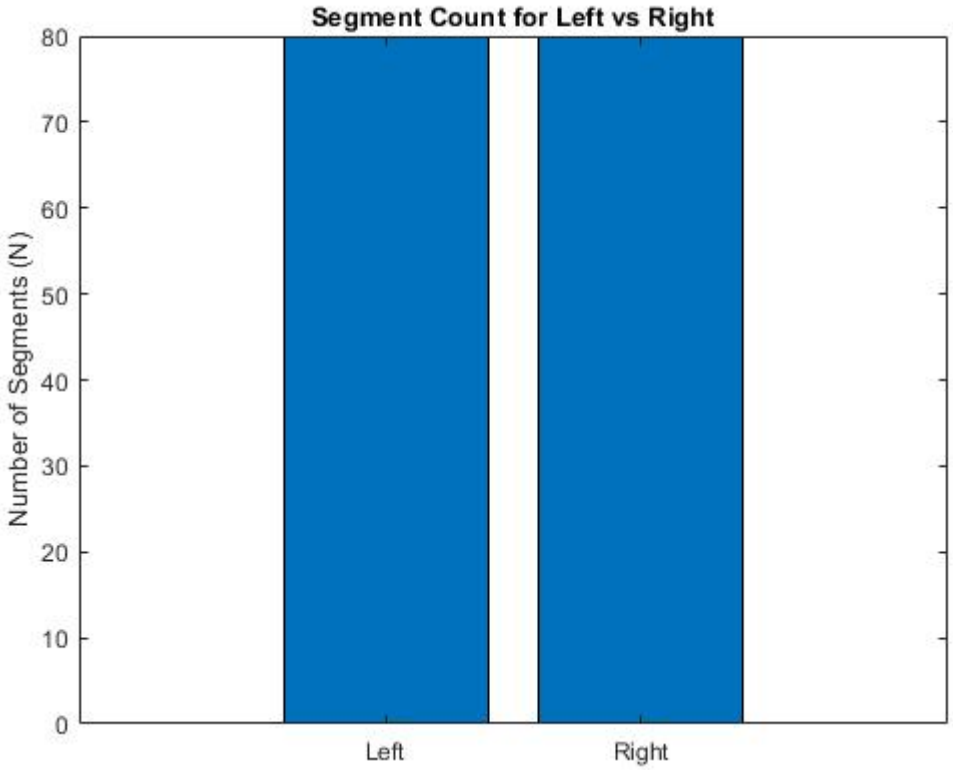


Figure 2. trials separated into 2 parts

Task 2: Average power

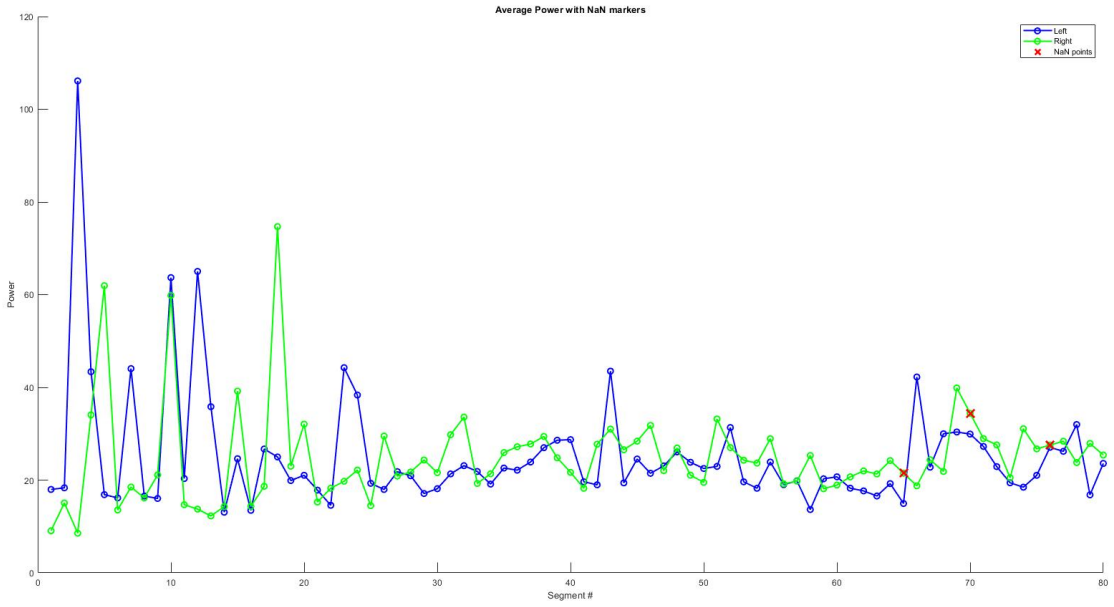


Figure 3. mean power of each electrode

Figure 3 shows the average power of each electrode during left (blue) versus right (green) imagined movement. Since there are multiple abnormal values, I filtered the result with a threshold, removing values greater than 1000 and replacing them with NaN (shown as red cross in the figure). Apparently, there is no clear gap between the two, the mean power of these two categories fluctuate around 20 for most of the trials.

Task 3: Separate the EEG response and calculate FFT

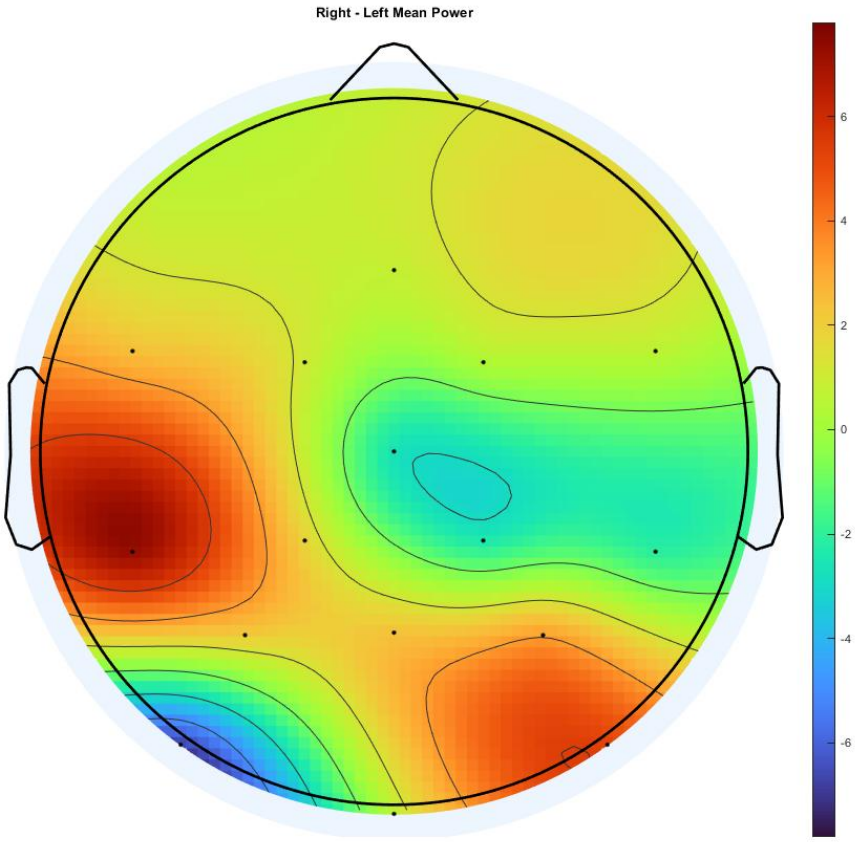


Figure 4. scalp maps of relative power (right - left)

According to Figures 4, relative power (right - left) is greater at the left part of the brain. This pattern occurs because right-hand motor imagery primarily activates the contralateral (left) sensorimotor cortex. As a result, power modulation is stronger over the left hemisphere, leading to higher relative power values (right − left) in that region.